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AI FOR AMERICANS FIRST

AI Protectionism, Energy and Semiconductors: US/Europe Divergence Trajectories 2024-2030

Integrated Geostrategic and Economic Analysis - Chapter II

76.9% global operational AI compute = USA	**1.59x energy cost EU/US**	**3.46:1 CACI ratio US/EU (Power Mode)**

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CHAPTER II

Methodology

2.1 General approach: mixed-method multi-scenario prospective analysis

This study combines a retrospective empirical analysis (2020-2026 diagnosis) with a prospective scenario-based projection (2026-2030). This two-pronged architecture responds to the nature of the phenomenon under investigation: AI technology protectionism is simultaneously an observable fact (export controls, Section 232 tariffs) and an ongoing process whose future trajectory depends on discretionary political variables, European strategic responses and partially unpredictable technological developments.

The retrospective component employs a descriptive quantitative method, based on the aggregation and cross-referencing of data from institutional sources (IEA, SIA/WSTS, Eurostat, EIA), industry reports (McKinsey, Deloitte, Epoch AI) and regulatory documents (Federal Register, BIS, White House). The objective is to establish a rigorous, sourced factual foundation covering three dimensions: data center energy consumption, the semiconductor market, and installed AI compute capacity by region.

The prospective component draws on the scenario planning tradition as formalized by Schwartz (1991) and practiced at Royal Dutch/Shell since the 1970s.[1] This method, belonging to the Intuitive Logics school (Bradfield et al., 2005), consists of constructing plausible and internally consistent scenarios - not to predict the future, but to explore the space of possibilities and evaluate the robustness of different strategies against divergent environmental developments.[2] It is particularly suited to situations characterized by high political and technological uncertainty, where classical econometric models reach their limits, which is precisely the case with AI technology protectionism.

Justification of the methodological choice. Three reasons underpin the choice of scenario methods over pure econometric modelling. First, the key analytical variables are largely political and discretionary: the decision of a US president to impose or not impose GPU quotas on Europe cannot be modelled through a regression function. Second, the interactions between the energy, technological and geopolitical dimensions are non-linear and systemic: a restriction on GPUs can, through cascade effects, alter energy investment flows, data center location decisions and the competitive structure of entire sectors. Third, the available data on installed compute by region are partial and heterogeneous: no unified public database of AI FLOPs by country exists, making rigorous econometric calibration premature.

The chosen method therefore combines the quantitative rigour of the empirical diagnostic (sourced data, time series, measurable ratios) with the qualitative flexibility of scenario construction, in the spirit of what Schoemaker (1995) calls a disciplined heuristic.[3] The scenarios are not probabilistic forecasts but coherent strategic narratives, each based on explicit assumptions and developing their consequences through measurable metrics.

2.2 Data sources: classification and critical evaluation

The study draws on three categories of sources, whose reliability and potential biases must be explicitly acknowledged. This methodological transparency conforms to the recommendations of the OECD/JRC Handbook on Constructing Composite Indicators (Nardo et al., 2008), which prescribes systematic documentation of sources, their limitations and their biases in any composite indicator construction.[4]

2.2.1 Primary sources (official and regulatory documents)

This category includes normative or institutional texts: presidential proclamations (Section 232), BIS rules (AI Diffusion Rule, Entity List), IEA reports, European Parliament publications (EPRS), official statistical data (SIA/WSTS for semiconductors, Eurostat and EIA for energy, RTE for France). These sources offer the highest factual reliability but may contain institutional framing biases: the IEA tends to favour moderate scenarios, the European Parliament to emphasize EU sovereignty risks.

2.2.2 Academic sources and think tanks

This category includes peer-reviewed articles (Farrell and Newman, 2019; Bresnahan and Trajtenberg, 1995; Brynjolfsson et al., 2019; Mugge, 2024) and publications from recognized think tanks (Bruegel, Carnegie Endowment, CSIS, OECD, Federal Reserve Board). The former provide robust theoretical grounding; the latter offer empirically founded policy analyses but are potentially influenced by each institution ideological orientation. We prioritize cross-referencing sources of different orientations (Bruegel / Carnegie / Fed) to limit this bias.

2.2.3 Industry and consulting sources

McKinsey, Deloitte, Accenture, Epoch AI and CFG Europe provide market data, sectoral projections and capacity estimates unavailable in public sources. These sources present a potential systematic bias: consulting firms have an interest in amplifying trends (to justify transformation engagements) and market estimates are often optimistic. We mitigate this bias by triangulating figures with institutional data and explicitly flagging discrepancies between sources. For example, 2024 semiconductor sales are 627.6 billion USD according to the SIA (traditional scope) but 775 billion USD according to McKinsey (expanded scope), a 24 percent gap reflecting methodological differences, not inconsistencies.[5]

2.2.4 AI compute data: the Epoch AI GPU Clusters dataset

Measuring installed AI compute by country constitutes the central methodological challenge of this study. We rely primarily on the Epoch AI GPU Clusters dataset (Pilz, Rahman, Sanders and Heim, 2025), which catalogues over 500 supercomputers and GPU clusters worldwide for the period 2019-2025.[6] This dataset, available under an open Creative Commons Attribution licence, constitutes the most comprehensive and systematically documented source on global AI compute infrastructure to date. It is used as a reference by the Stanford AI Index Report (2025), by several government reports, and by institutions such as OpenAI and DeepMind.

The dataset covers for each cluster: country of location, chip type (H100, A100, GB200, TPU, etc.), computational performance in 16-bit FLOP/s, number of H100 equivalents, operational date, sector (private/public), electrical power (MW) and estimated hardware cost. This granularity enables aggregation by country and year, directly meeting the needs of our variable F(r) in the CACI.

Limitations of the Epoch AI dataset. Three limitations must be highlighted. First, coverage is estimated at 10-20 percent of global aggregate AI compute performance (March 2025), with significant heterogeneity across companies and chip types: approximately 20-37 percent of NVIDIA H100s, 12 percent of A100s, but less than 4 percent of Google TPUs and a negligible fraction of custom chips from AWS, Microsoft or Meta.[7] Second, Chinese systems are anonymized (names removed, values rounded to one significant figure), limiting analytical precision for China. Third, the physical location of a cluster does not determine access: many clusters are accessible via cloud services from other countries.

We complement these data with the OECD Working Paper by Lehdonvirta, Wu, Hawkins et al. (October 2025), which develops a methodology for estimating public cloud AI compute availability by country, counting cloud regions from major providers equipped with AI accelerators (A100, H100, GB200) across 39 economies.[8] This complementary approach distinguishes installed compute (Epoch AI) from accessible compute (OECD), a distinction crucial for the CACI.

2.3 Scenario construction

Scenario construction follows a four-step protocol, inspired by the 2x2 matrix methodology (Schwartz, 1991; van der Heijden, 2004) and adapted to the geostrategic context of AI.[9]

Step 1 - Identification of driving forces. Predetermined elements (whose evolution is reasonably predictable regardless of scenario) include: continued growth in global AI compute demand; structural increase in data center energy consumption; European dependence on Asian and American foundries for leading-edge chips; concentration of the global cloud around three US hyperscalers; and exponential increase in frontier model training costs.

Critical uncertainties (whose evolution depends on political choices, strategic reactions or technological disruptions) are grouped along two dimensions. Dimension 1 - Intensity of US technology protectionism: this dimension covers a spectrum from maintaining current restrictions to aggressive hardening (GPU quotas for the EU, restrictions on APIs and models, explicit prioritization of deliveries to US companies). Dimension 2 - European response capacity: this dimension covers a spectrum from passive posture (marginal adaptation, acceptance of dependence) to active response (Compute Zones with derogated energy, accelerated AI Factories, SMR nuclear for data centers, alternative partnerships Japan-Korea-Taiwan, AI Act revision).

Step 2 - 2x2 matrix and scenario generation. Crossing the two uncertainty dimensions generates a four-scenario matrix. The choice of four rather than three scenarios is deliberate. Schwartz (1991) and Shell method practitioners recommend never building three scenarios, as the human mind tends to treat the middle scenario as the most likely, thereby reducing the exercise utility.[10] The 2x2 matrix forces the analyst to explore extreme quadrants, precisely where strategic ruptures play out.

Step 3 - Narrative development and quantification. Each scenario is developed following a standardized protocol comprising three components: a strategic narrative describing the plausible sequence of events between 2026 and 2030; a quantification of key metrics calibrated on 2020-2026 empirical data and projected according to the scenario assumptions; and early warning indicators (leading indicators) enabling identification, from 2026-2027 onward, of which scenario reality is converging toward.

Step 4 - Sensitivity analysis and robustness. For each recommendation formulated in Chapter VII, we evaluate its robustness across all four scenarios. A recommendation is considered robust if it produces positive or neutral results in at least three of the four scenarios.

2.4 Key metrics and original indicator: the CACI

We define six metrics to be calculated or estimated in the empirical diagnostic (Chapter III), then projected in each scenario (Chapter V). Together, they form a dashboard of US/EU divergence in AI: M1 (compute gap, US/EU installed AI FLOPs ratio normalized by GDP), M2 (relative FLOP cost for training), M3 (cloud dependence, share of EU AI workloads on US infrastructure), M4 (sectoral AI productivity), M5 (energy constraint, data center demand/capacity ratio) and M6 (AI relocations).

2.4.1 Theoretical foundations of the CACI

Grounding in the literature. The construction of a compute-centred AI competitiveness composite indicator responds to a need identified by several converging research streams. Since 2023-2024, academic and institutional literature increasingly emphasizes that computational capacity has become the most discriminating factor of production for frontier AI (Sevilla et al., 2022; Epoch AI, 2025; Pilz et al., 2025). US export controls (BIS, October 2022; updated 2023 and 2025) explicitly place advanced compute at the heart of geopolitical competition, while Hawkins, Lehdonvirta and Wu (2025) introduce the concept of compute sovereignty as a structuring dimension of strategic autonomy.[11]

Yet existing AI competitiveness indices do not place compute at the centre of their construction. The IMF AI Preparedness Index (Cazzaniga et al., 2024), covering 174 countries, aggregates four dimensions (digital infrastructure, human capital, innovation/economic integration, regulation/ethics) without directly measuring installed compute capacity.[12] The Tortoise Media Global AI Index (2024), ranking 83 countries on 122 indicators across three pillars (Implementation, Innovation, Investment), includes an infrastructure/supercomputing component but subsumes it within a weighted additive index where compute is merely one factor among many.[13] The Stanford AI Index (2025) provides rich compute trend data but proposes no composite index. None of these instruments formalizes the mechanism by which compute, adjusted for its energy cost and related to an economy absorptive capacity, determines AI competitiveness.

The CACI aims to fill this gap by proposing a parsimonious yet theoretically grounded indicator that captures the multiplicative interaction between four factors: accessible compute, the relevant labor force, geopolitical access conditions and the energy cost.

2.4.2 Formal definition

The CACI for a region r at period t is defined in two complementary modes. In Power Mode (absolute): CACI(r,t) = F(r,t)^0.40 x L(r,t)^0.20 x R(r,t)^0.15 / E(r,t)^0.25. In Intensity Mode (per unit of national wealth): CACI(r,t) = F(r,t)^0.40 x L(r,t)^0.20 x R(r,t)^0.15 / (E(r,t)^0.25 x GDP(r,t)).

F(r,t) - installed and accessible AI compute capacity in region r at period t. We aggregate the Epoch AI GPU Clusters dataset by country, in H100-equivalents (the metric used by Epoch AI), and complement it with OECD estimates of accessible cloud compute. F admits two specifications used in parallel: F_total (all clusters located in the territory regardless of ownership, defining the Physical CACI) and F_dom (clusters owned and operated by domestic actors, defining the Sovereign CACI).

L(r,t) - working population with AI competencies (proxy: STEM graduates plus certified AI training plus LinkedIn Economic Graph AI skill density). This factor captures absorptive capacity in the sense of Cohen and Levinthal (1990): abundant compute without human capital to exploit it does not produce competitiveness.[14]

R(r,t) - geopolitical access index (between 0 and 1) capturing the country position in the export-control regime. R = 1 for the United States (unlimited access to frontier compute), R = 0.95 for Tier-1 close allies (UK), R = 0.9 for the EU and France (Tier-1 with selective restrictions), R = 0.85 for India (Tier-2), R = 0.7 for China (heavily restricted).

E(r,t) - average effective energy cost of compute in region r, in USD per MWh, adjusted for hyperscaler Power Purchase Agreements. The dashboard reference values for the April 2026 snapshot are USA 85, China 92, France 115, Germany 140, UK 190 USD per MWh. These figures correspond to PPA-adjusted prices effectively faced by data centers, not to raw Eurostat industrial tariffs (which are typically higher in Europe). EU effective costs in this PPA-adjusted scope are 1.4 to 1.7 times US levels.[15]

GDP(r,t) - gross domestic product of region r (World Bank, Eurostat). Used only in Intensity Mode to normalise across economies of very different sizes.

Justification of the multiplicative form. The choice of multiplicative (geometric) rather than additive (arithmetic) aggregation is deliberate and rests on three arguments. First, General Purpose Technology theory (Bresnahan and Trajtenberg, 1995) posits strong complementarity between innovation inputs: compute without affordable energy or qualified human capital produces no competitiveness gains, justifying a form where weakness in one factor penalizes the whole. Second, the OECD/JRC Handbook (2008, p. 33) recommends geometric aggregation when components are not perfectly substitutable: unlike the arithmetic mean, the geometric mean does not allow a very high score on one dimension to fully compensate a very low score on another. Third, recent work on AI index construction (Koronakos, Kritikos and Sotiros, 2024; analysis of the Tortoise GAII via Choquet integral) confirms that AI competitiveness dimensions exhibit interactions (complementarities and redundancies) that make simple linear aggregation problematic.[16]

Economic interpretation. In logarithmic form, the CACI in Intensity Mode can be written ln(CACI) = 0.40 ln(F) + 0.20 ln(L) + 0.15 ln(R) - 0.25 ln(E) - ln(GDP). This transformation linearizes the relationship and allows each coefficient to be interpreted as an elasticity. The indicator is designed for bilateral comparison: the ratio CACI(US)/CACI(EU) measures relative competitive advantage. The April 2026 snapshot yields a ratio of 3.46:1 in Power Mode (F_total) - this is the headline figure of this study.

2.4.3 Calibration protocol and data sources

CACI calibration follows a four-step protocol consistent with OECD/JRC Handbook (2008) recommendations: identification and collection of raw data; treatment of missing values and normalization; aggregation; and sensitivity analysis.

F(r,t) - Installed AI compute capacity. Estimation follows a bottom-up approach. The main layer comes from the Epoch AI GPU Clusters dataset (April 2026 version), providing 16-bit FLOP/s and H100-equivalent performance aggregated by country for 2019-2026. National shares of operational compute on the April 2026 snapshot are: USA 76.9 percent, China 12.8 percent, EU (13 main economies) 4.4 percent, Norway 1.1 percent, Japan 1.0 percent.[17] When all reported clusters are included (operational plus planned), the USA share moves to about 50 percent, with the Gulf (UAE, Saudi Arabia) and Korea capturing a large share of the projected 2027-2030 buildout. The complementary layer comes from the OECD (Lehdonvirta et al., 2025), cataloguing cloud regions with AI accelerators across 39 economies, capturing the accessibility dimension that physically installed compute does not fully reflect.

F aggregation procedure. For each country-year, we extract from the Epoch AI dataset the sum of H100-equivalents from all clusters located in the country, filtering for confirmed systems (certainty greater than or equal to Likely); apply an extrapolation factor to correct for the dataset under-coverage (10 to 20 percent of global compute), using Epoch AI sectoral coverage estimates by chip type; and add estimated accessible cloud capacities via the OECD methodology for countries where foreign cloud compute represents a significant share of effective capacity. Raw data and calculation code (Python/pandas) are documented in the methodological appendix and on the public dashboard.[18]

E(r,t) - Energy cost. The dashboard CSV reference values used throughout this study, in USD per MWh and PPA-adjusted, are: USA 85, China 92, France 115, Germany 140, UK 190, India 88. These figures incorporate negotiated large-consumer tariffs (hyperscaler PPAs), using IEA (2025, Energy and AI) estimates of data center energy mix by region. They are systematically lower than raw Eurostat industrial tariffs in Europe, because hyperscaler PPAs and direct nuclear/renewable sourcing materially reduce the effective cost faced by AI data centers. The Federal Reserve Board (October 2025) documents a significant negative correlation between energy costs and AI adoption at the European firm level.[19]

GDP(r,t) and L(r,t). GDP is available from the World Bank (World Development Indicators) and Eurostat. The dashboard April 2026 snapshot uses GDP values in trillion USD: USA 29.3, China 18.7, EU 18.9, France 3.16, Germany 4.68, UK 3.6. The AI human capital proxy L(r,t) combines three sub-indicators: number of STEM graduates (OECD Education at a Glance), AI skills density as measured by LinkedIn Economic Graph (profiles with AI skills relative to working population) and AI certifications (estimates based on certified AWS/Google/Microsoft cloud programs by country). The dashboard snapshot uses, in millions of AI-skilled workers: USA 3.5, China 4.8, EU 3.1, France 1.5, Germany 1.9, UK 1.8. We acknowledge a bias favouring English-speaking countries and economies where LinkedIn is dominant, and document this bias effect in the sensitivity analysis (section 2.4.5).[20]

CACI normalization and benchmark selection. The raw CACI produces heterogeneous absolute values whose direct interpretation is challenging. In accordance with standard practice (Nardo et al., 2008; Saisana and Tarantola, 2002), we apply a min-max normalization to the leader: each country normalised CACI is expressed as a percentage of the United States raw CACI, so that CACI_norm(USA) = 100 by construction.

This normalization choice is methodologically grounded on three arguments. First, the United States constitutes a natural benchmark given its dominant position across all four CACI factors: 76.9 percent of global operational AI compute (Epoch AI, April 2026), the lowest effective industrial energy costs among major OECD economies (USD 85 per MWh, PPA-adjusted, versus USD 115 in France and USD 140 in Germany - dashboard CSV, April 2026), the highest nominal GDP, and an unmatched AI human capital ecosystem. Second, normalization to the leader is the established convention for relative competitiveness indices: Balassa RCA (1965), the WEF GCI (Schwab, 2019) and UNIDO Competitive Industrial Performance Index all proceed by ratio to the sectoral or national leader. Third, this normalization eliminates interpretive ambiguities of absolute values and enables temporal comparisons: a France CACI moving from 25 to 35 between 2024 and 2028 signifies partial catch-up, even if both countries raw values have increased.

The most important methodological consequence of this normalization is the revelation of a structural gap that traditional indices conceal. The IMF AI Preparedness Index (Cazzaniga et al., 2024) assigns scores of 0.85 (USA) and 0.74 (France), a ratio of 1.15:1, suggesting near-parity. The Tortoise Global AI Index produces similar gaps. Yet the normalized CACI in Power Mode places France at about 25 and Germany at about 5 on a scale where the United States = 100, i.e. ratios of 4:1 and 19:1 respectively. The headline US/EU ratio (EU aggregated as 13 main economies) is 3.46:1. This divergence is not contradictory: it results from the fact that multi-dimensional indices linearly aggregate normative dimensions (regulation, ethics, publications) that arithmetically compensate for deficits on material factors (compute, energy). The CACI, by isolating these material factors, exposes the physical chasm that composite metrics average away.

This property of the CACI - revealing the true magnitude of the compute gap - constitutes the central analytical contribution of the index and the fundamental empirical thesis of this study. The CACI gap is not a normalization artifact: it is the scientific result.

2.4.4 Positioning relative to existing AI competitiveness indices

The IMF AI Preparedness Index (AIPI) covers 174 countries (2023) and aggregates four pillars: digital infrastructure, human capital and labour market policies, innovation and economic integration, regulation and ethics. The AIPI does not measure installed AI compute and does not weight by energy cost. Its expected correlation with the CACI is positive but imperfect: countries scoring highly on the AIPI (Singapore, Denmark, Netherlands) are not necessarily those with the most effective compute per unit of GDP.[21]

The Tortoise Media Global AI Index (GAII) ranks 83 countries on 122 indicators across three pillars (Implementation, Innovation, Investment). It includes an infrastructure/supercomputing component but aggregates it linearly with subjective weights (acknowledged by Tortoise as a limitation). The GAII is broader and more multidimensional than the CACI, but precisely because it is broad, it dilutes the compute signal in a many-indicator composite. As Koronakos et al. (2024) show, the GAII weighting subjectivity can invert country rankings depending on chosen weighting scenarios.

The Stanford AI Index (2025) constitutes the most comprehensive reference in terms of raw data (notable models by country, investment, publications, patents, compute trends). It does not propose a composite index but provides the time series used by many other indices. The Stanford AI Index public data are available via an open Google Drive folder.[22]

CACI specific value added. The CACI distinguishes itself through four properties: it places compute at the centre rather than the periphery of the indicator, reflecting compute now-dominant role in frontier AI; it explicitly integrates energy cost as a bottleneck, consistent with IEA (2025) findings; it uses a theoretically grounded multiplicative aggregation rather than additive; and it is parsimonious (four variables), making it transparent and reproducible, at the cost of lesser comprehensiveness.

2.4.5 CACI limitations and sensitivity analysis

First limitation - opacity of $F(r,t)$. Installed compute measurement depends on incomplete private data. The Epoch AI dataset covers only 10-20 percent of global compute, with uneven coverage across sectors and companies. National attribution is itself debatable: a growing share of compute is held by a few private hyperscalers operating globally. Mitigation: we systematically document uncertainty margins, present ranges rather than point values, and verify result stability when F varies by plus or minus 30 percent. The Physical/Sovereign decomposition partially addresses the attribution problem.

Second limitation - qualitative heterogeneity of compute. The CACI aggregates FLOPs without distinguishing GPU generations (an H200 does not equal an A100 in energy efficiency and real performance). Mitigation: the Epoch AI dataset provides performance in H100 equivalents, offering partial normalization. We propose a GPU generation weighting factor in the appendix and show that adjustment marginally modifies rankings.

Third limitation - the human capital proxy L(r,t). The STEM plus LinkedIn plus certifications combination presents a bias favouring English-speaking countries. This bias likely underestimates China and some Asian economies. Mitigation: we replicate the analysis using the IMF AIPI Human Capital sub-index as an alternative proxy and show ranking sensitivity to this choice.

Fourth limitation - indicator staticity. The CACI measures a state at a given moment, not a dynamic. This is why we calculate it across multiple years (2022, 2024, 2026) and project it in each scenario, enabling trajectory tracking and gap evolution assessment.

Fifth limitation - endogeneity. Countries gaining AI productivity invest massively in compute, creating a reverse causality risk: the CACI might capture the consequence rather than the cause of competitiveness. This study, within its dissertation framework, does not formally instrument this relationship (no instrumental variables or GMM strategy). However, we note that the exogenous shock of the October 2022 BIS rules offers a natural quasi-experiment that could ground a causal identification strategy in Difference-in-Differences. We identify the formal treatment of this endogeneity (DiD, IV or Arellano-Bond GMM) as a priority research avenue for any publishable extension of this work.[23]

Sixth limitation - normalization bias for small economies. Comparative analysis of the CACI against the IMF AIPI and Tortoise GAII indices reveals an instructive anomaly: certain countries with low GDP and a reduced AI workforce obtain disproportionately high CACI scores. South Africa is emblematic: ranked last or second-to-last on the IMF AIPI and Tortoise indices, it can appear as a global leader on the Intensity Mode CACI when the GDP normalisation amplifies a small-denominator effect. This phenomenon is a classic case of normalization bias identified in the composite indicator literature. The OECD/JRC Handbook (Nardo et al., 2008, pp. 27-29) warns that normalization by GDP or population can produce misleading results for small economies. Our mitigation is threefold: a critical mass threshold on F (excluding economies below 10 000 H100-equivalents); a scaling factor alpha(r) = min(1, F(r) / F_median) for sensitivity tests; and a dual ranking (Power Mode for absolute power, Intensity Mode for density), with the explicit caveat that bilateral comparisons in this study focus on USA, EU, France and China.[24]

2.4.6 Reproducible numbers - April 2026 snapshot

To allow the reader to replay the calculation from the public dashboard CSVs, we expose here the inputs and outputs used throughout this study. The four input variables are sourced from the dashboard data folder (energy_prices.csv, gdp_data.csv, workforce_data.csv, gpu_clusters.csv) at https://mo0ogly.github.io/America-First-IA/dashboard/data/.

Inputs (April 2026 snapshot). USA: F_total = 39.65 million H100-eq, L = 3.5 million, R = 1.00, E = 85 USD/MWh, GDP = 29.3 trillion USD. EU (13 main economies aggregated): F_total = 2.62 million H100-eq, L = 3.1 million, R = 0.90, E = 135 USD/MWh, GDP = 18.9 trillion USD. France: F_total = 2.44 million H100-eq, L = 1.5 million, R = 0.90, E = 115 USD/MWh, GDP = 3.16 trillion USD. Germany: F_total = 0.05 million H100-eq, L = 1.9 million, R = 0.90, E = 140 USD/MWh, GDP = 4.68 trillion USD. China: F_total = 0.40 million H100-eq, L = 4.8 million, R = 0.70, E = 92 USD/MWh, GDP = 18.7 trillion USD.

Outputs (Power Mode F_total, normalised at USA = 100). USA = 100.0; EU(13) = 28.9; France = 25.3; India = 22.2; China = 15.7; UK = 7.0; Germany = 5.4. UAE = 55.7 in Physical CACI but only 6.0 in Sovereign CACI (F_dom restricted to clusters owned by domestic entities G42, MGX, Mubadala, Khazna, TII), illustrating the Physical/Sovereign decomposition: 99.6 percent of UAE F_total is owned by US-side actors (Stargate UAE, Microsoft, OpenAI). Headline ratios: US/EU(13) = 3.46:1; US/France = 3.96:1; US/Germany = 18.59:1; US/China = 7.46:1.

These outputs are reproduced live on the public dashboard. Any subsequent dashboard update (new clusters, energy revisions, GDP refresh) will mechanically refresh the figures; the methodology does not change. The Power Mode F_total ratio of 3.46:1 reported on the cover and throughout this study corresponds exactly to this snapshot.

2.5 Scope and delimitations

Geographic scope. The analysis focuses on the bilateral United States / European Union relationship, with a specific focus on France. China is treated as a contextual variable (primary target of US export controls, factor of pressure on chip production capacities) but is not subject to in-depth analysis. Japan, South Korea and Taiwan appear as semiconductor supply chain actors.

Temporal scope. The diagnostic covers 2020-2026, scenarios cover 2026-2030. The 2030 horizon is chosen as it corresponds to the convergence of several deadlines: IEA projections for data center energy, expected maturity of the EU Chips Act, France 2030 SNIA objectives and potential arrival of the first operational SMR nuclear reactors.

Technological scope. The study covers frontier AI (foundation models, compute-intensive) and its material prerequisites (GPU/ASIC, data centers, energy). It integrates AI robotics as an amplifying factor for energy demand. It does not address edge embedded AI (smartphones, IoT), except insofar as it constitutes a specific objective of the French SNIA.

2.6 General methodological limitations

Radical political uncertainty. Technology protectionism depends on discretionary political decisions with structurally low predictability. A change of US administration in 2028, an unexpected US-EU trade agreement, or an escalation of the US-China conflict could invalidate certain assumptions. This is precisely why we propose four scenarios rather than a single trajectory.

Technological disruptions. The DeepSeek episode (January 2025), where a Chinese model achieved near-frontier performance with substantially reduced training budget, illustrates the possibility of efficiency breakthroughs that would alter the problem terms. The IEA (2025, Energy and AI) devotes a case study to DeepSeek and concludes that even with significant efficiency improvements, demand growth absorbs gains (Jevons rebound effect).[25]

Compute data opacity. The exact number of GPUs deployed per hyperscaler, the precise geographic distribution of data centers and GPU volumes exported by region are partially or fully confidential data. Our installed compute estimates carry significant margins of error, which we systematically document.

Consulting source bias. As noted in section 2.2, industry sources carry a systematic optimism bias. We mitigate this through triangulation but cannot eliminate it entirely.

These limitations do not compromise the analysis validity. The scenario method is precisely designed to function in high-uncertainty environments, where the objective is not prediction but structured exploration of possibilities. As Schwartz notes, scenarios are not forecasts; they are plausible stories that help you think.[26] Our contribution lies in the rigour of the framing, the explicitness of assumptions, the transparency of data sources and the originality of the CACI indicator, rather than in the precision of numerical projections.

Notes

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13. Tortoise Media (2024), The Global Artificial Intelligence Index 2024. See also Koronakos, G., Kritikos, M. and Sotiros, D. (2024), 'Mitigating Subjectivity and Bias in AI Development Indices,' Expert Systems with Applications.

14. Cohen, W.M. and Levinthal, D.A. (1990), 'Absorptive Capacity: A New Perspective on Learning and Innovation,' Administrative Science Quarterly, 35(1), pp. 128-152.

15. Effective costs are derived from the dashboard energy_prices.csv and reflect PPA-adjusted prices. Raw Eurostat industrial tariffs in Europe (consumption band IE) can be 1.5 to 2 times higher; the gap between raw and effective prices is documented in Chapter III. The Federal Reserve Board (October 2025) documents a significant negative correlation between energy costs and AI adoption at the European firm level.

16. Geometric aggregation is also used by the UNDP Human Development Index since 2010, for analogous reasons of non-substitutability between dimensions. See OECD/JRC Handbook (2008), pp. 31-33.

17. Operational shares computed by aggregating Epoch AI clusters with status Existing/Operational/Online/In progress on the April 2026 snapshot.

18. The Python code and data are reproducible. The Epoch AI dataset is downloadable in CSV at https://epoch.ai/data/gpu_clusters.csv (daily refresh). The dashboard CSVs are mirrored at https://mo0ogly.github.io/America-First-IA/dashboard/data/.

19. The LinkedIn bias is documented: in countries where LinkedIn is rarely used (China, Russia, some Southeast Asian economies), AI skills density is mechanically underestimated.

20. IMF (2024), AI Preparedness Index Dashboard. Singapore, Denmark, Netherlands and the US occupy top positions; China ranks 31st (score 0.63).

21. Maslej, N. et al. (2025), 'Artificial Intelligence Index Report 2025,' Stanford Institute for Human-Centered AI, April 2025.

22. Tortoise Media (2024), op. cit. Stanford AI Index public data: open Google Drive folder.

23. The most promising causal identification strategy would exploit the exogenous shock of the October 2022 BIS rules in a Difference-in-Differences framework. See Arellano, M. and Bond, S. (1991), 'Some Tests of Specification for Panel Data,' Review of Economic Studies, 58(2), pp. 277-297.

24. OECD/JRC Handbook (2008), pp. 27-29. The dual ranking solution (intensity / scale) is also used by Tortoise Media (2024) in the Global AI Index. Saisana, M. and Tarantola, S. (2002), State-of-the-art Report on Current Methodologies and Practices for Composite Indicator Development, JRC European Commission, pp. 14-16.

25. IEA (2025), Energy and AI, devotes a case study to DeepSeek and concludes that even with significant efficiency improvements, demand growth absorbs gains (Jevons rebound effect).

26. Schwartz (1991), op. cit., p. 38. Author translation.

Public dashboard: https://mo0ogly.github.io/America-First-IA/dashboard/

Repository: https://github.com/mo0ogly/America-First-IA